

Problem 1 (Source: ???)

Let x, y , and z are positive real numbers such that

$$x^{\log_2(yz)} = 2^8 \cdot 3^4, \quad y^{\log_2(zx)} = 2^9 \cdot 3^6, \quad \text{and} \quad z^{\log_2(xy)} = 2^5 \cdot 3^{10}.$$

Compute the smallest possible value of xyz .

Solution. Solution

□

Problem 2 (Source: ???)

Let $\lfloor z \rfloor$ denote the greatest integer less than or equal to z . Compute

$$\sum_{j=-1000}^{1000} \left\lfloor \frac{2025}{j+0.5} \right\rfloor.$$

Solution. Solution

□

Problem 3 (Source: ???)

Let a , b , and c be real numbers such that $a^2(b+c) = 1$, $b^2(c+a) = 2$, and $c^2(a+b) = 5$. Find the minimum possible value of abc .

Solution. Adding the three equations gives us that

$$a^2b + ab^2 + a^2c + ac^2 + b^2c + bc^2 = 8.$$

We then add $2abc$ and factor the LHS to get $(a+b)(b+c)(c+a) = 8 + 2abc$.

Multiplying the three equations gives

$$a^2(b+c)b^2(c+a)c^2(a+b) = 10.$$

The LHS can be rewritten as $(abc)^2(a+b)(b+c)(c+a) = (abc)^2(8 + 2abc)$ via our substitution, so $(abc)^2(8 + 2abc) = 10$.

We now let $n = abc$. We then get that $n^2(8 + 2n) = 10$ or $2n^3 + 8n^2 - 10 = 0$, which has real roots 1 and $\frac{-5 \pm \sqrt{5}}{2}$, and the minimum of these roots is $\boxed{\frac{-5 - \sqrt{5}}{2}}$.

We now show that this is the minimum value of abc . **[Incomplete Solution]** □

Problem 4 (Source: ???)

Real numbers a, b, c, d satisfy $ab = cd$ and

$$a^2 + b^2 + c^2 + d^2 = 1.$$

Compute the greatest possible value of $29a + 11b + 61c + 19d$.

Solution. Solution

□